Lecture 15 - Optimization I: The First Derivative Test

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Updates

Grading taking longer than expected, my apologies!
Review Session: 4:30-6:30 in Vincent Hall 16 (the auditorium in the basement of Vincent).
This place is HUGE so feel free to invite your friends in other sections of 1271 to come.
Today's lecture content will ***NOT*** be tested on Exam 2, it is Exam 3 content.

Quiz Time!

Let’s get into neat rows, spread out, and slay this quiz!

Optimization

Derivative Review

Today's content arguably the *most important* application of the derivative, as such let's review some key facts about the derivative.

1. The *first* derivative of $f(x)$ at $x = x_0$ is the slope of the best linear approximation (linearization) at $x = x_0$:
$$\mathscr{L}_{x=x_0}(x) = f(x_0) + f'(x_0)(x - x_0) \tag{1}$$
2. As such, the *first* derivative is related to increasingness/decreasingness by: $f(x)$ increasing/decreasing at $x = x_0 \iff f'(x_0)$ is positive/negative.
3. Curvature of $f(x)$ at $x = x_0$ is controlled by the sign of the *second* derivative $f''(x_0)$: concave up/down $\iff f''(x_0)$ positive/negative
4. Comes from *best quadratic approximation*:
$$\mathscr{Q}_{x=x_0}(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 \tag{2}$$

Towards Optimization

After modelling some real world relationship as a function $output = f(input)$, one often wants to find the perfect input quantity that maximizes/minimizes the output quantity.
E.g.:

- If the relationship is: input=characteristic of material, output=cost of producing material with that characteristic then companies often want to *minimize*
- If the relationship is: input=cost of R&D and output=net profit from R&D companies often want to *maximize*
- If input=pH of substrate and output=reaction rate then both maxima and minima tell you important information about the chemical process.

Optimization Defns. I

To mathematically model this type of behavior we need to lay down a few defns.

Global Minima/Maxima/Extrema

Given a function $f(x)$, we say that an *output value* $f(c)$ is a *global* minimum/maximum resp. if, for any other input x_0 in the domain of $f(x)$, $f(c)$ is $\leq$ / $\geq$ resp. than $f(x_0)$.
If $f(c)$ is *either* a global minimum/global maximum we call it a global *extremum*.

This is fantastic for real world applications, but derivatives struggle to classify these because the derivative is something defined locally only to a point. This (hopefully) motivates us to adapt the above definition to something more *local*.

Optimization Defns. II

Local Minima/Maxima/Extrema

Given a function $f(x)$, we say that an *input/output pair* $(c, f(c))$ is a *global* minimum/maximum resp. if, there is some small neighborhood $(c-\epsilon, c+\epsilon)$ around c, for which any $c-\epsilon < x_0 < c+\epsilon$, $f(c)$ is $\leq$ / $\geq$ resp. than $f(x_0)$.
If $(c, f(c))$ is *either* a local minimum/local maximum we call it a local *extremum*.

Given how similar the defns. are, these two concepts are very tightly related:

Global Extrema are Local Extrema

If $f(c)$ is a global minimum/maximum/extremum then $(c, f(c))$ is a local minimum/maximum/extremum.

Optimization Defns. Example

Let's differentiate (pun maybe intended) the two defns. by examining the wavy function $f(x) = \frac{\sin(x)}{x}$:

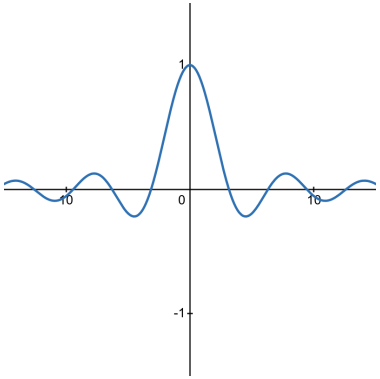


Figure: Graph of $\frac{\sin(x)}{x}$

Optimization I

If we want to use calculus tools to help us find local/global extrema, we should first figure out what tools of ours actually contain enough information to detect local/global extrema in the first place.

In English: which of our approximations actually themselves have local/global extrema that are *related* to those of the original function?

Attempt 1: Linearization. Do linearizations have local/global extrema, can they recognize those in the original function?

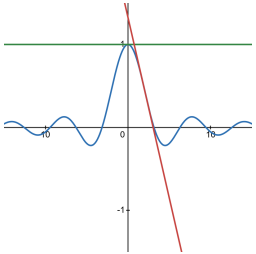


Figure: Graph of $\frac{\sin(x)}{x}$ with two linearizations graphed.

Optimization II

Lines suck! Everything or nothing is an extrema, which makes us sad. Let's bump it up to quadratic approximations.

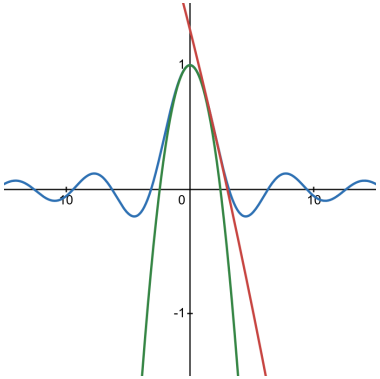


Figure: Graph of $\frac{\sin(x)}{x}$ with two quadratic approximations graphed.

Optimization III

Quadratic approximations detect extrema! The extrema for parabola occurs at their vertex, and the vertex of $f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$ is at $(x_0, f(x_0))$ if and only if $f'(x_0) = 0$! If this seems weird, let's take what we've learned and reinterpret it in a much more direct manner:

1. We needed to take a quadratic approximation at $x = x_0$, this means that $f(x)$ has what property? $f(x)$ is second differentiable at $x = x_0$.
2. The quantity $f'(x_0)$ was important; if $f(x)$ is second differentiable at $x = x_0$ what does that imply about $f'(x)$ at $x = x_0$? $f(x)$ is continuously differentiable at $x = x_0$!
3. Can we rephrase the condition being a local minimum/maximum in this setting to relate it to conditions on the derivative? Yes!

Optimization IV

Let's assume that $f(x)$ is a function that is both continuously differentiable at $x = x_0$ and that has a local extrema at $(x_0, f(x_0))$.
If we have a local *minimum* at $x = x_0$ then:

- If $x_0 - \epsilon < x \leq x_0$ (to the left of x_0) then what can we say about $f(x_0)$, $f(x)$ and $f'(x_0)$? Minimality implies $f(x_0) < f(x)$ and so by mean value theorem, $f'(x) \leq 0$ for $x_0 - \epsilon < x < x_0$.
- If $x_0 \leq x < x_0 + \epsilon$ (to the right of x_0) then what can we say? Minimality implies $f(x_0) < f(x)$ so by MVT $f'(x) \geq 0$.
- If $f'(x) \leq 0$ for $x < x_0$, $f'(x) \geq 0$ for $x > x_0$ *AND* $f'(x)$ is continuous at $x = x_0$, then that forces $f'(x_0) = 0$!

Moral of the story: At a minimum, we start decreasing (negative derivative) then increase (positive derivative) which forces the derivative at the switching point $f'(x_0)$ to be zero.
Exercise: think about what happens when instead $(x_0, f(x_0))$ is a local *maximum*.

Optimization V

We can compile this into the following important result:

First Derivative Test

If $(x_0, f(x_0))$ is a local extremum (or global extremum) and $f(x)$ is continuously differentiable at $x = x_0$, then we *must* have $f'(x_0) = 0$.

Unfortunately, this does not *perfectly* classify local extrema, even for the nicest (infinitely differentiable) functions: consider $f(x) = x^3$:

Optimization VI

This condition is so important we give it its own name:

Critical Point

For a function $f(x)$ we say that x_0 is a *critical point* of $f(x)$ if $f'(x_0)$ is 0 or for technical reasons if $f'(x_0)$ is *undefined*.

The undefined condition is to handle functions like $f(x) = \sqrt[3]{x^2}$:

Optimization VII

Why do we care then?
Recall that solving equations like $f'(x) = 0$ often has us end up with a *finite* set of points and searching over a *finite* set of points is MUUUCH easier than, for instance, searching over all of $\mathbb{R}$ (super duper infinite).
This gives us the following procedure: in order to find the extrema of a function $f(x)$ we can avoid the futile task of comparing all the infinite values $f(c)$ for $c \in \mathbb{R}$ by instead:

- Showing/recognizing that $f(x)$ is continuously differentiable
- Solving for all the critical points of $f(x)$
- Computing $f(c)$ for every critical point c
- Comparing those above values to find the largest (maximum) and smallest (minimum).

Optimization Example I

Let's optimize the function $f(x) = \frac{x^5}{5} - \frac{x^4}{4} - \frac{2x^3}{3} + 2$!
We first find the critical points:

$$f'(x) = x^4 - x^3 + 2x^2 \tag{3}$$
$$= x^2(x^2 - x + 2) \tag{4}$$
$$= x^2(x-2)(x+1) \tag{5}$$

So we have critical points at $x = 0, 2, -1$.
But are they extrema? I.e. in addition to having $f'(c) = 0$ does $f'(c)$ actually switch signs at $x = c$?

Optimization Example II

This is where the notion of a sign graph comes in:

Sign Graph

Given a function $f(x)$, the sign graph of $f(x)$ is a useful way to depict only the positivity or negativity of a function.
We start with drawing out the domain of $f(x)$ as a number line, mark each x value where $f(x) = 0$ or is undefined, and then (assuming $f(x)$ is continuous) can characterize the sign over each split interval just by evaluating at a single point.

Optimization Example III

Thus we have that $f'(x)$ only switches signs across $x = 2$ and $x = -1$.
This means that:

1. $x = 0$ is just a critical point, and not an extrema.
2. $x = 2$ is where the sign switches from negative to positive, so $x = 2$ is a local minimum.
3. $x = -1$ is where the sign switches from positive to negative so $x = -1$ is a local maximum.

Thus we've determined that, for $f(x) = \frac{x^5}{5} - \frac{x^4}{4} + \frac{2x^3}{3} + 2$, there is a local minimum at $x = 2$ and a local maximum at $x = -1$. Check the graph, and we see that's correct!

Optimization Example IV

Let's instead consider $f(x) = x + \sqrt[3]{x^2}$.
We start by finding critical points:

$$f'(x) = 1 + \frac{2}{3\sqrt[3]{x}} \tag{6}$$

$$= \frac{2 + 3\sqrt[3]{x}}{3\sqrt[3]{x}} \tag{7}$$

Which is zero when:

$$2 + \sqrt[3]{x} = 0 \tag{8}$$

$$\sqrt[3]{x} = -2 \tag{9}$$

$$x = (-2)^3 = -8 \tag{10}$$

and is undefined when:

$$3\sqrt[3]{x} = 0 \tag{11}$$

$$\sqrt[3]{x} = 0 \tag{12}$$

$$x = 0^3 = 0 \tag{13}$$

Optimization Example V

We then construct the sign graph of $f'(x) = \dfrac{2 + 3\sqrt[3]{x}}{3\sqrt[3]{x}}$.

Which gives us that at $x = -8$ the sign changes from positive to negative,
and at $x = 0$ the sign changes from negative to positive.
Therefore $x = -8$ is a local maximum, and $x = 0$ is a local minimum.

Worksheet Time

Let's get into groups and work on the worksheet!

End of Class

That’s all folks!